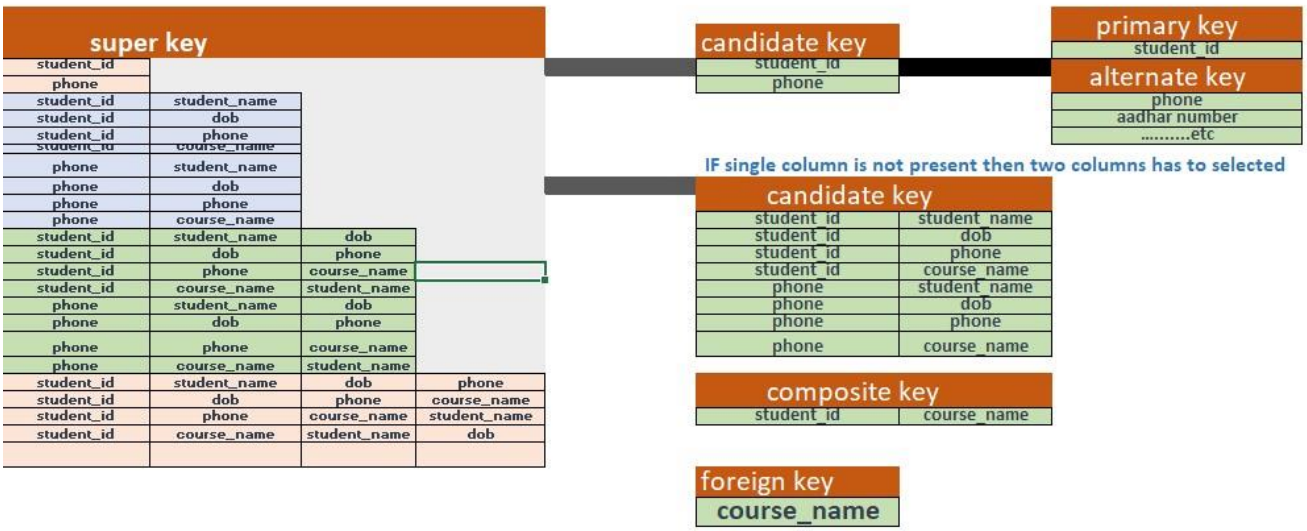
Key

The attribute / attributes which uniquely identifies the record

1. Super key
   The attribute or set of attributes which uniquely identifies the record
2. Candidate key
   The minimal subset of super key
3. Primary key
   An attribute which uniquely identifies the record and present in Candidate key list
4. Alternate key
   Except primary key attribute remaining attributes among Candidate key
5. Composite key
   If primary key has more than one attribute
6. Foreign key
   Attribute which builds relationship between the tables

Example

student_id	student_name	dob	phone	course_name
101	Smith	03-Apr-00	9234567890	python
102	Allen	09-Jul-01	8234567890	sql
103	Clark	05-Dec-99	8734567890	web
104	James	06-Apr-00	6334567890	devops
105	Clark	03-Apr-00	9934567890	web



NORMALIZATION

Normalization is the process of organizing data in a database

- Which is helpful to protect the data
- To make the database more flexible by eliminating redundancy and inconsistent dependency.
- To overcome the anomalies like (insertion,updation,deletion)

First normal form (1NF)

- Table should have exactly one primary key
- Columns should have unique name
- Each cell in a table must have atomic value
- Values in a column must be of same domain

Second normal form (2NF)

- Table should be in first normal form
- Table should not have partial functional dependencies

Third normal form (3NF)

- Table should be in second normal form
- Table should not have transitive functional dependencies

Boyce codd normal form (BCNF or 3.5NF)

- Table should be in third normal form
- If Column A is determining Column B then Column A should be primary key / super key / candidate

PARTIAL FUNCTIONAL DEPENDENCIES

If a table consist of composite key(primary key with more than one attribute) and non-prime attributes are depending on part of the primary key then table has partial functional dependencies

----solution

It has to be removed by dividing table into sub tables

EXAMPLE

student_id	student_name	dob	phone	course_name	fees	course_duration
101	Smith	03-Apr-00	9234567890	python	5500	60 days
102	Allen	09-Jul-01	8234567890	sql	4500	35 days
103	Clark	05-Dec-99	8734567890	web	5000	40 days
104	James	06-Apr-00	6334567890	devops	5000	35 days
105	Clark	03-Apr-00	9934567890	web	5000	40 days
102	Allen	09-Jul-01	8234567890	python	5500	60 days

student_id	student_name	dob	phone
101	Smith	03-Apr-00	9234567890
102	Allen	09-Jul-01	8234567890
103	Clark	05-Dec-99	8734567890
104	James	06-Apr-00	6334567890
105	Clark	03-Apr-00	9934567890

course_id	course_name	fees	course_duration
10	python	21-Jan-15	60 days
20	sql	26-Apr-12	35 days
30	web	08-Sep-13	40 days
40	devops	08-Sep-13	35 days

student_id	course_id
101	10
102	20
103	30
104	40
105	30
102	10

TRANSITIVE FUNCTIONAL DEPENDENCIES

If non-prime attributes are determining (identifying) another non-prime attributes then it is known as transitive functional dependencies

----solution

It has to be removed by dividing table into sub tables

EXAMPLE

course_name	fees	course_duration	Room_no	trainer_id	trainer_name	login	logout
python	5500	60 days	1	t1	Mr Irshan	8:00 AM	4:00 PM
sql	4500	35 days	2	t2	Ms Sneha	9:00 AM	5:00 PM
web	5000	40 days	3	t3	Mr yalguresh	10:00 AM	6:00 PM
devops	5000	35 days	1	t1	Mr Irshan	8:00 AM	4:00 PM
django	7000	65 days	4	t4	Mr Rahul Raj	10:00 AM	6:00 PM
Data Structure	0	35 days	4	t4	Mr Rahul Raj	10:00 AM	6:00 PM

trainer_id	trainer_name	login	logout
t1	Mr Irshan	8:00 AM	4:00 PM
t2	Ms Sneha	9:00 AM	5:00 PM
t3	Mr yalguresh	10:00 AM	6:00 PM
t4	Mr Rahul Raj	10:00 AM	6:00 PM

course_id	course_name	fees	course_duration	trainer_id
10	python	5500	60 days	t1
20	sql	4500	35 days	t2
30	web	5000	40 days	t3
40	devops	5000	35 days	t1
50	django	7000	65 days	t4
60	Data Structure		35 days	t4